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basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

SENIOR CERTIFICATE EXAMINATIONS/ NATIONAL SENIOR CERTIFICATE EXAMINATIONS

LIFE SCIENCES P1

2023

MARKING GUIDELINES

MARKS: 150

These marking guidelines consist of 9 pages.

PRINCIPLES RELATED TO MARKING LIFE SCIENCES

1. **If more information than marks allocated is given**
Stop marking when maximum marks is reached and put a wavy line and 'max' in the right-hand margin.
2. **If, for example, three reasons are required and five are given**
Mark the first three irrespective of whether all or some are correct/ incorrect.
3. **If whole process is given when only a part of it is required**
Read all and credit the relevant part.
4. **If comparisons are asked for but descriptions are given**
Accept if the differences/similarities are clear.
5. **If tabulation is required but paragraphs are given**
Candidates will lose marks for not tabulating.
6. **If diagrams are given with annotations when descriptions are required**
Candidates will lose marks.
7. **If flow charts are given instead of descriptions**
Candidates will lose marks.
8. **If sequence is muddled and links do not make sense**
Where sequence and links are correct, credit. Where sequence and links are incorrect, do not credit. If sequence and links become correct again, resume credit.
9. **Non-recognised abbreviations**
Accept if first defined in answer. If not defined, do not credit the unrecognised abbreviation but credit the rest of the answer if correct.
10. **Wrong numbering**
If answer fits into the correct sequence of questions but the wrong number is given, it is acceptable.
11. **If language used changes the intended meaning**
Do not accept.
12. **Spelling errors**
If recognisable, accept the answer, provided it does not mean something else in Life Sciences or if it is out of context.
13. **If common names are given in terminology**
Accept, provided it was accepted at the national memo discussion meeting.
14. **If only the letter is asked for but only the name is given (and vice versa)**
Do not credit.

15. If units are not given in measurements

Candidates will lose marks. Memorandum will allocate marks for units separately.

16. Be sensitive to the sense of an answer, which may be stated in a different way.

17. Caption

All illustrations (diagrams, graphs, tables, etc.) must have a caption.

18. Code-switching of official languages (terms and concepts)

A single word or two that appear(s) in any official language other than the learners' assessment language used to the greatest extent in his/her answers should be credited if it is correct. A marker that is proficient in the relevant official language should be consulted. This is applicable to all official languages.

19. Changes to the memorandum

No changes must be made to the memoranda without consulting the provincial internal moderator who in turn will consult with the national internal moderator (and the Umalusi moderators where necessary).

20. Official memoranda

Only memoranda bearing the signatures of the national internal moderator and the Umalusi moderators and distributed by the National Department of Basic Education via the provinces must be used.

SECTION A**QUESTION 1**

1.1	1.1.1	D✓✓		
	1.1.2	B✓✓		
	1.1.3	A✓✓		
	1.1.4	B✓✓		
	1.1.5	B✓✓		
	1.1.6	A✓✓		
	1.1.7	C✓✓		
	1.1.8	B✓✓		
	1.1.9	C✓✓		
	1.1.10	B✓✓	(10 x 2)	(20)
1.2	1.2.1	Homeostasis✓		
	1.2.2	Mitochondria✓		
	1.2.3	Alzheimer's✓ disease /dementia		
	1.2.4	Choroid✓		
	1.2.5	Precocial✓ development		
	1.2.6	Islets of Langerhans✓		
	1.2.7	Acrosome✓		
	1.2.8	Umbilical artery✓	(8 x 1)	(8)
1.3	1.3.1	Both A and B✓✓		
	1.3.2	None✓✓		
	1.3.3	B only✓✓	(3 x 2)	(6)
1.4	1.4.1	(a) Peripheral✓ nervous system		(1)
		(b) Autonomic nervous system✓		(1)
	1.4.2	Spinal✓ nerves		(1)
	1.4.3	E✓ - Parasympathetic nervous system✓		(2)
	1.4.4	Neurons✓		(1)
	1.4.5	- Meninges✓ - Cranium✓/bone tissue - Cerebrospinal fluid✓	Any	(2)
		(Mark first TWO only)		(8)
1.5	1.5.1	Semi-circular canals✓		(1)
	1.5.2	Ossicles✓		(1)
	1.5.3	(a) D✓ - Eustachian tube✓		(2)
		(b) C✓ - Oval window✓		(2)
	1.5.4	(a) Maculae✓		(1)
		(b) Cristae✓		(1)
				(8)

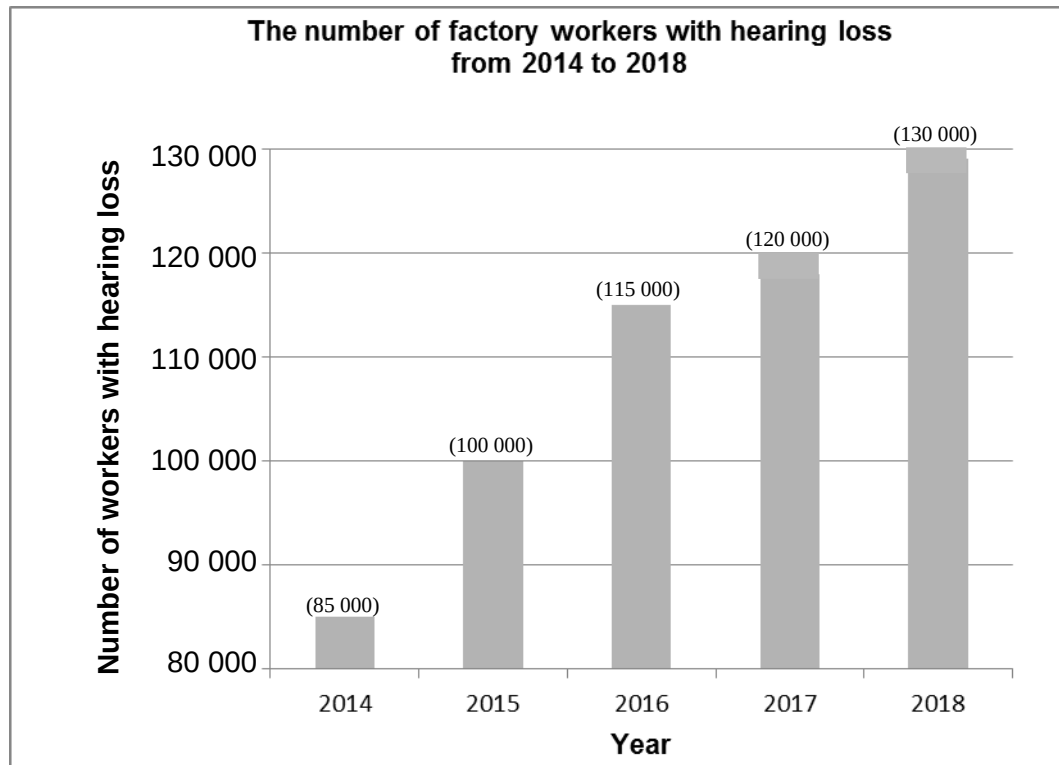
TOTAL SECTION A: 50

SECTION B**QUESTION 2**

2.1.1	Endometrium✓		(1)
2.1.2	Fertilisation✓		(1)
2.1.3	The (nucleus of the) sperm fuses with (the nucleus of) the ovum✓		(1)
2.1.4	- Zygote divides by mitosis✓ - to form a (solid) ball of cells✓ - called the morula✓ - which develops into a hollow ball of cells✓ - called the blastula✓/blastocyst	Any	(4)
2.1.5	- It is muscular✓ - to protect the foetus from mechanical injury✓/to allow for parturition/birth - It is flexible✓/can expand - to accommodate the growing foetus✓ - It is hollow✓ - to accommodate the growing foetus✓ - The thickened endometrium✓ - allows for implantation✓ /survival of the embryo (Mark first TWO only)	Any (2 x 2)	(4)
2.1.6	- The secretion is alkaline✓ which - neutralises the acidic conditions✓ of the vagina		(2)
(13)			
2.2	2.2.1	- Stimulates the development of ovarian follicles✓ - Initiates puberty✓ (Mark first ONE only)	Any (1)
2.2.2	- LH✓/Luteinising Hormone		(1)
2.2.3	- LH stimulates ovulation✓ - therefore, ovulation will not take place✓ - There will be no ovum to fertilise✓	Any	(2)
2.2.4	- A Graafian follicle is not formed✓ - Since the Graafian follicle secretes oestrogen✓ - oestrogen levels will be reduced✓ therefore - the endometrium will not develop✓/ thicken - and no implantation can take place✓ OR - There is no ovum produced✓/a Graafian follicle is not formed - Ovulation does not occur✓ - No fertilisation✓ occurs and - a zygote is not formed✓ - and no implantation can take place✓		(5)
(9)			

- 2.3
- Under the influence of testosterone✓
 - diploid cells✓/germinal epithelium cells
 - in the seminiferous tubules✓of the testis
 - undergo meiosis✓ to form
 - haploid sperm✓
- Any (4)
- 2.4
- 2.4.1 External✓ fertilisation (1)
- 2.4.2
- To increase the chances of fertilisation✓
 - since the gametes may be lost✓/not reach one another due to predation✓/water currents
- OR**
- To produce more zygotes✓/offspring
 - since many will be lost✓
 - because they are preyed on✓/washed away/dry out
- (3)
- 2.4.3 The embryos develop inside an egg, outside the female's body✓ (1)
- (5)**
- 2.5
- 2.5.1
- It has a duct✓
 - The secretion is released externally✓/not released into blood
 - It secretes sweat✓/It does not secrete a hormone
- Any (2)
- (Mark first TWO only)**
- 2.5.2
- They receive stimuli✓ from the environment
 - and convert them to nerve impulses✓
- (2)
- 2.5.3
- More blood flows to the surface of the skin✓ to allow more heat to be lost✓
- OR**
- More blood flows to the sweat glands✓ to increase the production of sweat✓
- (2)
- (6)**
- 2.6
- 2.6.1 Cochlea✓ (1)
- 2.6.2
- $$\left[\frac{(130\,000 - 85\,000)}{85\,000} \right] \times 100 = 52,94\%$$
- (3)
- 2.6.3
- More factories✓ were built increase in supply & demand
 - More workers✓ were employed
 - Extended exposure to loud sounds✓
 - Lack of precautionary measures✓
- Any (1)
- (Mark first ONE only)**
- 2.6.4
- The impulse will not be transmitted to the cerebrum✓
 - and will not be interpreted✓
- (2)

2.6.5

**Criteria for marking graph:**

Criteria	Mark allocation
Type: Bar graph is drawn (T)	1
Caption of the graph includes both variables (C)	1
Correct labels on X-axis and Y-axis (L)	1
Correct scale for Y-axis Equal width of bars and spaces (S)	1
Plotting: (P)	
1- 4 co-ordinates are plotted correctly	1
All 5 co-ordinates are plotted correctly	2

(6)
(13)
[50]

Histogram or line graph drawn

- Lose marks for type of graph and for scale

Transposed axes:

- Can get full credit if axes labels are also swapped and bars are horizontal
- If labels are *not* corresponding, then lose marks for labels and scale
- Check that the plotting is correct for the given labels

QUESTION 3

- 3.1 3.1.1 November✓ (1)
- 3.1.2 - The concentration of abscisic acid increases✓
 - To stimulate the abscission✓/falling of leaves
 - To prepare the tree for dormancy✓ (3)
- 3.1.3 - Less sunlight✓/ less water/ cold conditions therefore
 - Decreased photosynthesis✓/ reduced transpiration/ lower
 energy demand/ low growth rate Any (1 x 2) (2)
 (Mark first ONE only) (6)
- 3.2 3.2.1 - Auxins promote the development of roots✓
 - It brings about (general) root growth✓
 causing their downward✓growth/positive geotropism (3)
- 3.2.2 - In the stem, the auxins stimulate growth✓ on the lower side
 causing the stem to grow/bend upwards✓
 - In the root, the auxins inhibit growth✓ on the lower side
 causing the root to grow/bend downwards✓ (4)
 (7)
- 3.3 3.3.1 Adrenal✓ gland (1)
- 3.3.2 (a) Aldosterone level✓/ increased aldosterone level (1)
 (b) Blood pressure✓ (1)
- 3.3.3 - 1 688 volunteers were used✓
 - The procedure was done 4 times for each individual✓ (2)
 (Mark first TWO only)
- 3.3.4 - All factors should be kept constant✓/there should be only one
 independent variable
 to ensure the validity✓ of the investigation
 - Dietary factors✓/examples
 can also influence the blood pressure✓ (2 x 2) (4)
 (Mark first TWO only)
- 3.3.5 To compare the blood pressure before and after the administration
 of aldosterone✓✓ (2)
- 3.3.6 - The high aldosterone✓ level
 - will increase the permeability of the renal tubules✓ for salt
 - More salt will be reabsorbed✓ (3)
 (14)

3.4	- Adrenalin causes glycogen to be converted to glucose✓ which - increases the blood glucose level✓ - The breathing muscles are stimulated✓ - to increase the rate and depth of breathing✓ - The heart muscle is stimulated✓ - to pump faster✓ - There is also an increase in blood pressure✓ - increasing the transport of oxygen and glucose✓ - The rate of cellular respiration is increased✓	Any	(8)
3.5	3.5.1 Iris✓		(1)
	3.5.2 - Helps to maintain the shape of the eye✓ - Plays a role in refraction of light✓ - Allows the transmission of light✓ - Prevents desiccation✓ of structures in the eye - Holds the retina in position✓ - Nourishment✓ of the eye - Prevents mechanical injury✓ in the eye Any (Mark first TWO only)		(2)
	3.5.3 - Area B contains (a high concentration of) photoreceptors✓/ cones - Area C contains no photoreceptors✓/ no rods & cones		(2)
	3.5.4 Astigmatism✓		(1)
	3.5.5 - Because the lens will become cloudy✓/opaque - no/less light will enter the eye✓ - causing no sight ✓/weak sight		(3)
	3.5.6 - The ciliary muscle contracts✓ - The ciliary body moves closer to the lens✓ - The suspensory ligaments slacken✓ - Tension on the lens decreases✓ - The lens becomes more convex✓/rounded - Light rays are refracted more✓ - To focus the light on the retina✓	Any	(6) (15) [50]
TOTAL SECTION B:			100
GRAND TOTAL:			150